US012415527B2

(12) United States Patent
Ali

(10) Patent No.: US 12,415,527 B2
(45) Date of Patent: Sep. 16, 2025

(54) **RANGE SENSOR BASED CALIBRATION OF ROADSIDE CAMERAS IN ADAS APPLICATIONS**

(71) Applicant: **GM Global Technology Operations LLC**, Detroit, MI (US)

(72) Inventor: **Kamran Ali**, Troy, MI (US)

(73) Assignee: **GM GLOBAL TECHNOLOGY OPERATIONS LLC**, Detroit, MI (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 290 days.

(21) Appl. No.: **18/352,666**

(22) Filed: **Jul. 14, 2023**

(65) **Prior Publication Data**

US 2025/0018960 A1 Jan. 16, 2025

(51) **Int. Cl.**

B60W 50/00 (2006.01)
B60W 50/04 (2006.01)
B60W 50/06 (2006.01)
G06T 7/80 (2017.01)

(52) **U.S. Cl.**

CPC ***B60W 50/04*** (2013.01); ***B60W 50/06*** (2013.01); ***G06T 7/80*** (2017.01); *B60W 2050/0052* (2013.01); *B60W 2050/0088* (2013.01); *G06T 2207/10028* (2013.01)

(58) **Field of Classification Search**

CPC B60W 50/04; B60W 50/06; B60W 2050/0052; B60W 2050/0088; G06T 7/80; G06T 2207/10028; G08G 1/0116; G08G 1/04; H04N 17/002; G01B 21/02

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

12,137,681 B2	11/2024	Sibley et al.	
2017/0075355 A1	3/2017	Micks et al.	
2020/0025578 A1	1/2020	Wygant et al.	
2020/0365013 A1	11/2020	Simon et al.	
2021/0150230 A1	5/2021	Smolyanskiy et al.	
2021/0264773 A1	8/2021	Zhang	
2021/0366273 A1*	11/2021	Pan	B60W 60/00184

(Continued)

OTHER PUBLICATIONS

Andre Ibisch et al., Towards Autonomous Driving in a Parking Garage: Vehicle Localization and Tracking Using Environment-Embedded LIDAR Sensors, 2013 IEEE Intelligent Vehicles Symposium (IV), Jun. 2013, Gold Coast, Australia.

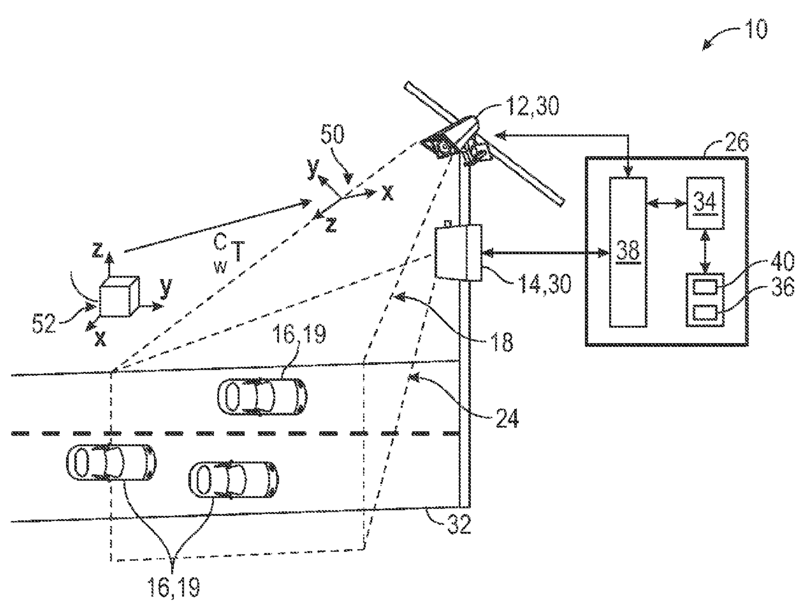
(Continued)

Primary Examiner — Ian Jen
(74) *Attorney, Agent, or Firm* — Vivacqua Crane, PLLC

(57) **ABSTRACT**

A system for range sensor-based calibration of roadside cameras includes one or more cameras and range sensors with overlapping fields of view of a roadway. Control logic stored within and executed by a controller includes control logic for capturing image and range sensor data, for filtering the data to focus on first and second regions of interest (ROIs); for filtering the data to focus on first and second movements of interest (MOIs); for filtering the data to focus on first and second objects of interest (OOIs); for filtering the data to focus on first and second positions of interest (POIs); and for defining that objects detected by the camera and sensor and that satisfy both the first and second ROI, MOI, OOI, and POI filters are matching objects. The control logic calibrates the camera by applying a correction factor based on the matching objects.

20 Claims, 3 Drawing Sheets



(56) **References Cited**

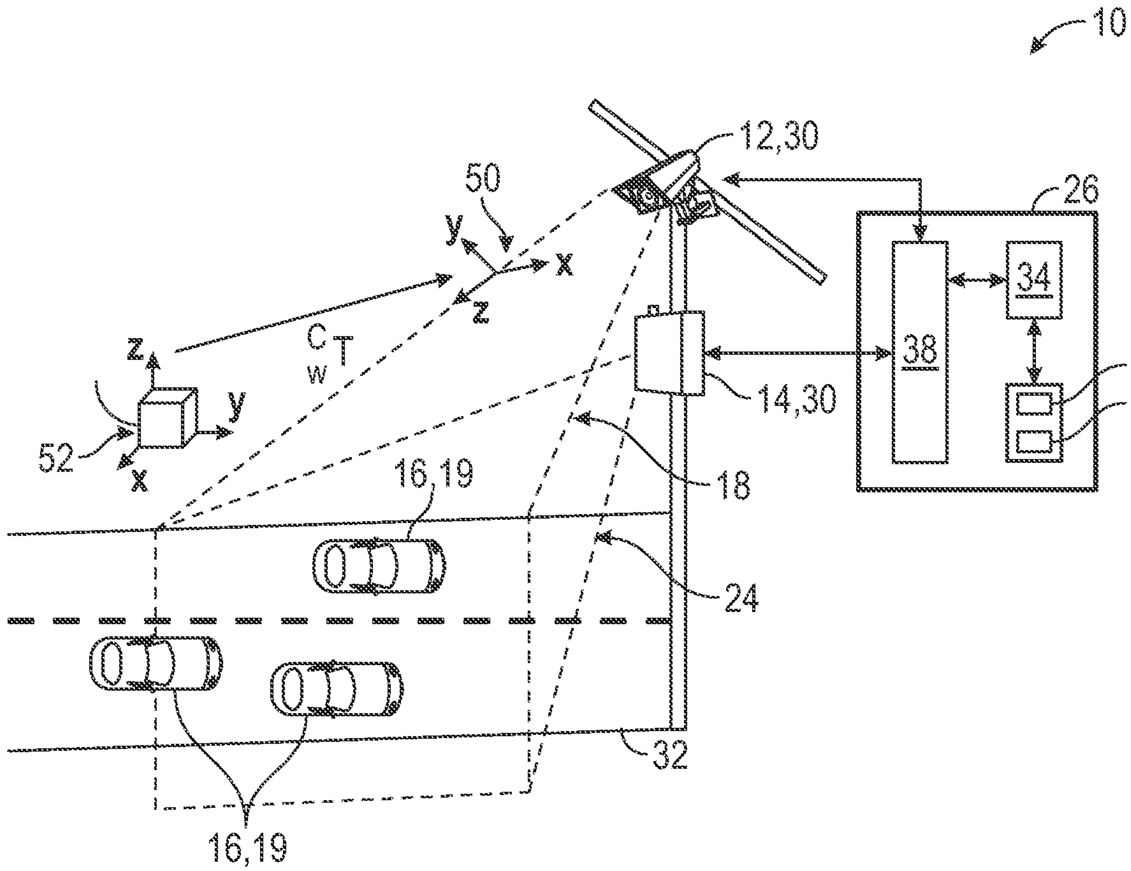
U.S. PATENT DOCUMENTS

2022/0266867	A1	8/2022	Ballard et al.
2024/0118104	A1	4/2024	Waschulzik

OTHER PUBLICATIONS

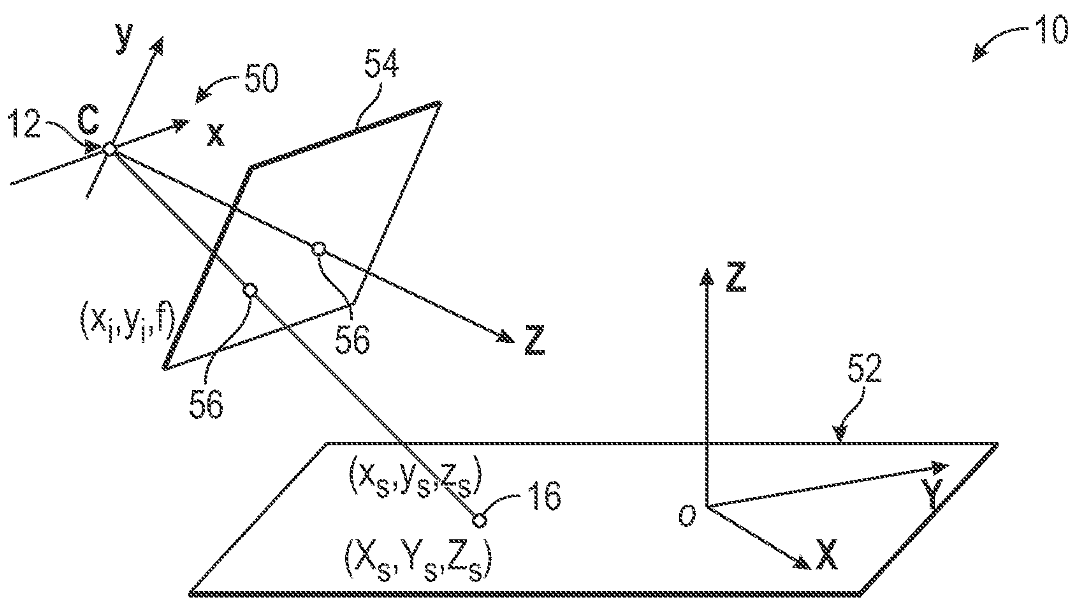
Byung Wook Kim et al., Vehicle Positioning Scheme Using V2V and V21 Visible Light Communications, 2016 IEEE 83rd Vehicular Technology Conference, May 2016.
Jens Einsiedler et al., External Visual Positioning System for Enclosed Carparks, 2014 11th Workshop on Positioning, Navigation and Communication, Mar. 2014.
Jens Einsiedler et al., Indoor Micro Navigation Utilizing Local Infrastructure-based Positioning, 2012 Intelligent Vehicles Symposium, Alcala de Henares, Spain, Jun. 2012.
United States Patent and Trademark Office. U.S. Appl. No. 17/739,417, filed May 9, 2022.
Du, Yuchuan, et al. A novel spatio-temporal synchronization method of roadside asynchronous MMW radar-camera for sensor fusion. IEEE transactions on intelligent transportation systems, 2021, 23. Jg., Nr. 11, S. 22278-22289. DOI: 10.1109/TITS.2021.3119079.
Caselitz, Tim, et al. "Monocular camera localization in 3d lidar maps." 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2016.
Cattaneo, Daniele, et al. "Cmrnet: Camera to lidar-map registration." 2019 IEEE intelligent transportation systems conference (ITSC). IEEE, 2019.
Guerrero-Ibanez, Juan, Sherali Zeadally, and Juan Contreras-Castillo. "Sensor technologies for intelligent transportation systems." Sensors 18.4 (2018): 1212.

* cited by examiner



10
12,30
50
26
y
x
z
34
z
C
w T
38
y
40
36
52
x
14,30
18
16,19
24
32
16,19

FIG. 1



10
y
50
54
12
C
x
Z
(x_i,y_i,f)
56
Z
56
52
(x_s,y_s,z_s)
16
o
Y
X
(X_s,Y_s,Z_s)

FIG. 2

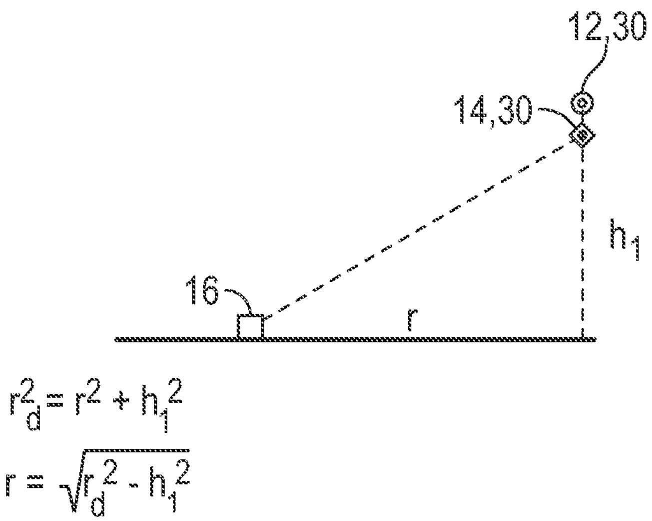

12,30
14,30
h_1
16
r
$r_d^2 = r^2 + h_1^2$
$r = \sqrt{r_d^2 - h_1^2}$

FIG. 3

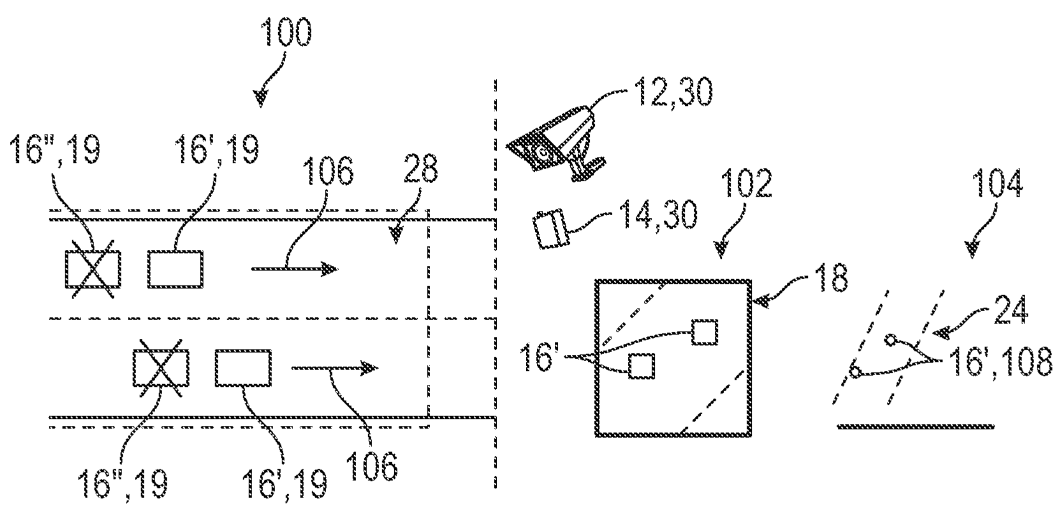

100
12,30
16",19
16',19
106
28
102
104
14,30
18
24
16'
16',108
16",19
16',19
106

FIG. 4

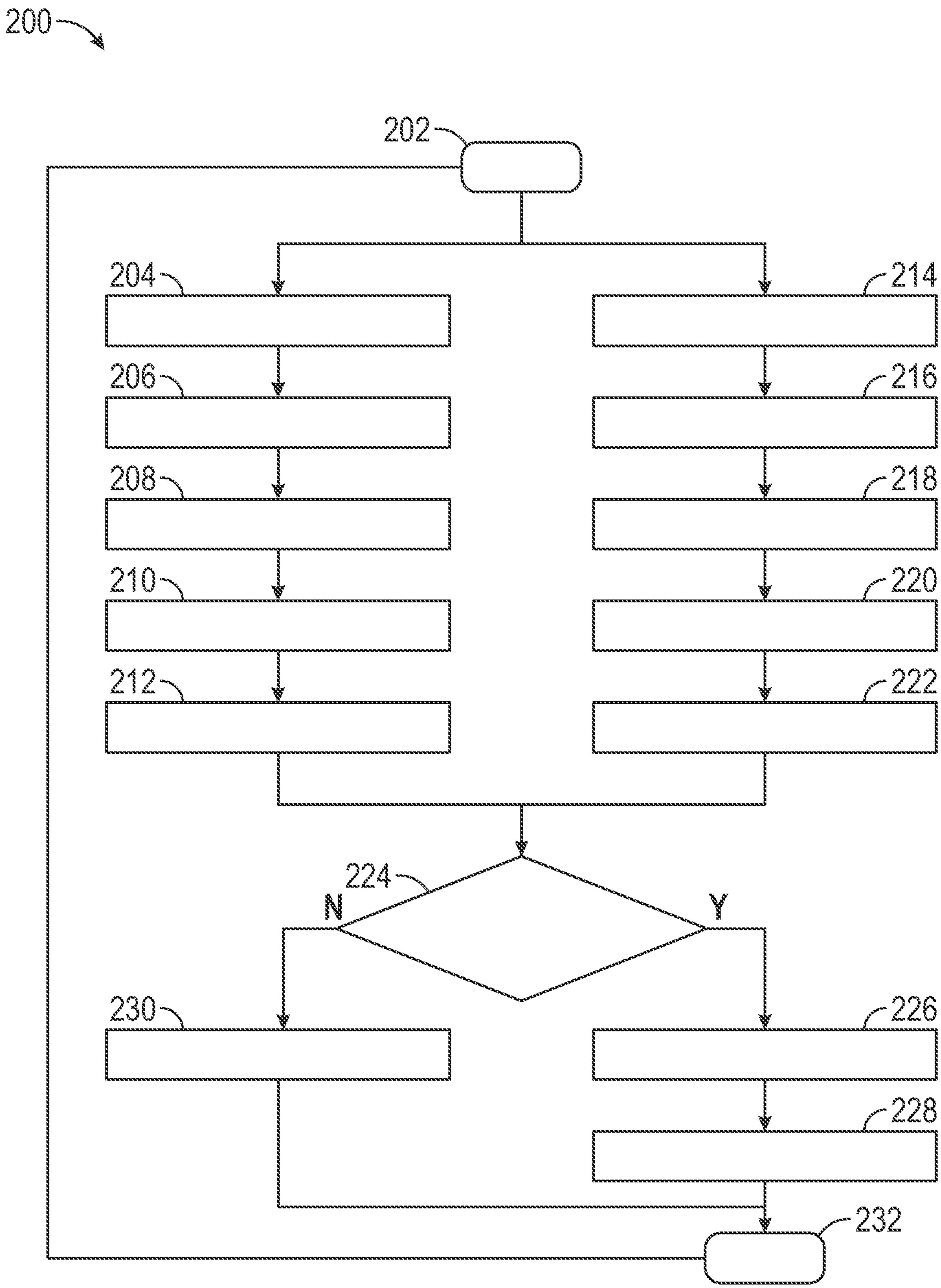

200
202
204
206
208
210
212
214
216
218
220
222
224
N
Y
230
226
228
232

FIG. 5

RANGE SENSOR BASED CALIBRATION OF ROADSIDE CAMERAS IN ADAS APPLICATIONS

INTRODUCTION

The present disclosure relates to infrastructure-based roadside cameras and the systems and methods of calibrating such roadside cameras.

Camera-based advanced driver assistance systems (ADAS) applications require calibration between roadside cameras and the real-world to perform position estimation in real-world coordinates. Typical calibration methodologies include manually surveying an area to determine real-world positions detectable by the roadside cameras being calibrated. Such manual survey and calibration procedures are time-consuming, cumbersome, and can be otherwise inefficient and complex.

Accordingly, while current systems and methods for calibrating roadside cameras are effective and achieve their intended purpose, there is a need for new and improved systems and methods for calibrating roadside cameras for use in ADAS applications that improve efficiency, decrease complexity, and reduce the quantities of time needed to perform calibration.

SUMMARY

According to several aspects, of the present disclosure a system for range sensor-based calibration of roadside cameras includes one or more roadside cameras. The roadside cameras are fixedly mounted to infrastructure and have a first field of view of at least a portion of a roadway. The system further includes one or more range sensors. The range sensors are fixedly mounted to infrastructure and have a second field of view of at least a portion of a roadway. The first and second fields of view at least partially overlap. The system further includes one or more controllers. Each of the one or more controllers has a processor, a memory, and one or more input/output (I/O) ports. The I/O ports are in communication with the one or more range sensors and the one or more roadside cameras. The memory stores programmatic control logic. The processor executes the programmatic control logic. The programmatic control logic includes at least a first, a second, a third, a fourth, a fifth, a sixth, a seventh, and an eighth control logic. The first control logic causes each of the one or more roadside cameras to capture image data, and causes each of the range sensors to capture range sensor data. The second control logic filters the image data to focus on a first region of interest (ROI), and filters the range sensor data to focus on a second ROI. The third control logic filters the image data to focus on a first movement of interest (MOI), and filters the range sensor data to focus on a second MOI. The fourth control logic filters the image data to focus on a first object of interest (OOI), and filters the range sensor data to focus on a second OOI. The fifth control logic filters the image data to focus on a first position of interest (POI), and filters the range sensor data to focus on a second POI. The sixth control logic selectively determines that objects detected by the roadside camera and by the range sensor satisfy both the first and second ROI, MOI, OOI, and POI filters and defines the objects that satisfy both the first and second ROI, MOI, OOI, and POI filters as matching objects. The seventh control logic saves the matching objects in the memory for calibration. The eighth control logic calibrates the roadside camera by applying a correction factor based on the matching

objects, so that the roadside camera accurately and precisely reports locations and movements of the matching objects for use in ADAS applications.

In another aspect of the present disclosure the second control logic further includes control logic that constrains the image data from the roadside camera to a sub-region or ROI within the first field of view in which the image data captured by the roadside camera is reliably accurate and precise, and control logic that constrains the range sensor data from the range sensor to a sub-region or ROI within the second field of view in which the range sensor data captured by the range sensor is reliably accurate and precise. The ROIs within the first and second fields of view are substantially identical.

In yet another aspect of the present disclosure the third control logic further includes control logic that constrains the image data to focus on the first MOI. The first MOI includes predictable and consistent motions of objects along the roadway. The third control logic further includes control logic that constrains the range sensor data to focus on the second MOI. The second MOI includes predictable and consistent motions of objects along the roadway.

In yet another aspect of the present disclosure the predictable and consistent motions of objects along the roadway further include motions in a predefined set of directions, linear or straight-line motions, and motions at predefined accelerations and velocities.

In yet another aspect of the present disclosure the fourth control logic further includes control logic that constrains the image data to define first OOIs by selecting objects that satisfy the first ROI filter, and the first MOI filter, and that are directly and unobstructedly viewable by the roadside camera. The fourth control logic further includes control logic that constrains the range sensor data to define second OOIs by selecting objects that satisfy the second ROI filter, and the second MOI filter, and that are directly and unobstructedly viewable by the range sensor.

In yet another aspect of the present disclosure the fifth control logic further includes control logic that constrains the image data by selecting only first OOIs that are locations within the first ROI having predefined fixed or mobile positions, and control logic that constrains the range sensor data by selecting only second OOIs that are at locations within the second ROI having predefined fixed or mobile positions. The predefined fixed or mobile positions include: locations of lane lines, positions of roadway signage, trees, traffic signals, telecommunications poles, fire hydrants, and/or curbs, and positions of movable or moving OOIs.

In yet another aspect of the present disclosure the sixth control logic further includes control logic that defines the objects as matching objects by comparing objects that satisfy both the first and second ROI, MOI, OOI, and POI filters and verifying one or more of the objects that satisfy both the first and second ROI, MOI, OOI, and POI filters is an identical object detected by both the roadside camera and the range sensor.

In yet another aspect of the present disclosure the eighth control logic further includes control logic that applies the correction factor to image data captured by the roadside camera. The correction factor is based on known physical and geometrical locations of the roadside camera relative to known physical and geometrical locations of the range sensor, and takes a point location in a real-world coordinate system and translates the point to a camera coordinate system according to a calibration matrix $K[I_3|0_3]$ based on a world-homography matrix "H":

1.

$$H = KR[I_3 \mid -\tilde{C}]$$

where R is a rotation and C takes from the world coordinate system to the center of projection of the camera coordinate system;

2.

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ z_s \\ 1 \end{bmatrix}$$

with $\alpha_x = fk_x$ and $\alpha_y = -fk_y$; and $x_{pix} = u'/w'$ and $y_{pix} = v'/w'$ so that:

3.

$$\begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = K[I_3 \mid 0_3]$$

where α_x and α_y are focal lengths in pixels and x_0 and y_0 are the coordinates of a center of the image in pixels, "s" is a skew parameter that accounts for roadside camera perspective angle and position, and $K[I_3|0_3]$ is the calibration matrix.

In yet another aspect of the present disclosure the calibration matrix $K[I_3|0_3]$ is further adjusted based on the known relative positions of the roadside camera and the range sensor, and based on a radial range or distance r_d between the roadside camera the object in the world coordinate system such that and a height h_1 of the roadside camera above ground at which the roadside camera is fixedly mounted, such that: $r=\sqrt{r_d^2-h_1^2}$.

In yet another aspect of the present disclosure a method for range sensor-based calibration of roadside cameras includes capturing image data with one or more roadside cameras, where the roadside cameras are fixedly mounted to infrastructure and have a first field of view of at least a portion of a roadway, and capturing range sensor data with one or more range sensors, where the range sensors are fixedly mounted to infrastructure and have a second field of view of at least a portion of a roadway. The first and second fields of view at least partially overlap. The method further includes executing programmatic control logic stored within a memory of one or more controllers in communication with the one or more roadside cameras and the one or more range sensors. Each of the one or more controllers has a processor, the memory, and one or more input/output (I/O) ports. The I/O ports are in communication with the one or more range sensors and the one or more roadside cameras. The processor executes the programmatic control logic, including: filtering the image data to focus on a first region of interest (ROI), and filtering the range sensor data to focus on a second ROI; filtering the image data to focus on a first movement of interest (MOI), and filtering the range sensor data to focus on a second MOI; filtering the image data to focus on a first object of interest (OOI), and filtering the range sensor data to focus on a second OOI; filtering the image data to focus on a first position of interest (POI), and filtering the range sensor data to focus on a second POI. The method further includes executing control logic for selectively determining that objects detected by the roadside camera and by the range sensor satisfy both the first and second ROI, MOI, OOI, and POI filters and defining the objects as matching objects. The method further includes executing control logic for saving the matching objects in the memory for calibration; and calibrating the roadside camera by applying a correction factor based on the matching objects, so that the roadside camera accurately and precisely reports locations and movements of the matching objects for use in ADAS applications.

In yet another aspect of the present disclosure the method includes constraining the image data from the roadside camera to a sub-region or ROI within the first field of view in which the image data captured by the roadside camera is reliably accurate and precise, and constraining the range sensor data from the range sensor to a sub-region or ROI within the second field of view in which the range sensor data captured by the range sensor is reliably accurate and precise. The ROIs within the first and second fields of view are substantially identical.

In yet another aspect of the present disclosure the method includes constraining the image data to focus on the first MOI. The first MOI includes predictable and consistent motions of objects along the roadway. The method further includes constraining the range sensor data to focus on the second MOI. The second MOI includes predictable and consistent motions of objects along the roadway.

In yet another aspect of the present disclosure constraining the image and sensor data further includes selecting predictable and consistent motions of objects along the roadway including motions in a predefined set of directions, linear or straight-line motions, and motions at predefined accelerations and velocities.

In yet another aspect of the present disclosure the method includes constraining the image data to define first OOIs by selecting objects that satisfy the first ROI filter, and the first MOI filter, and that are directly and unobstructedly viewable by the roadside camera; and constraining the range sensor data to define second OOIs by selecting objects that satisfy the second ROI filter, and the second MOI filter, and that are directly and unobstructedly viewable by the range sensor.

In yet another aspect of the present disclosure the method includes constraining the image data by selecting only first OOIs that are locations within the first ROI having predefined fixed or mobile positions, and constraining the range sensor data by selecting only second OOIs that are at locations within the second ROI having predefined fixed or mobile positions. The predefined fixed or mobile positions include: locations of lane lines, positions of roadway signage, trees, traffic signals, telecommunications poles, fire hydrants, and/or curbs, and positions of movable or moving OOIs.

In yet another aspect of the present disclosure the method includes defining the objects as matching objects by comparing objects that satisfy both the first and second ROI, MOI, OOI, and POI filters and verifying one or more of the objects that satisfy both the first and second ROI, MOI, OOI, and POI filters is an identical object detected by both the roadside camera and the range sensor.

In yet another aspect of the present disclosure the method includes applying the correction factor to image data captured by the roadside camera. The correction factor is based on known physical and geometrical locations of the roadside camera relative to known physical and geometrical locations of the range sensor, and takes a point location in a real-world coordinate system and translates the point to a camera coordinate system according to a calibration matrix $K[I_3|0_3]$ based on a world-homography matrix "H":

1.

$$H = KR[I_3 \,|\, -\tilde{C}]$$

where R is a rotation and C takes from the world coordinate system to the center of projection of the camera coordinate system;

2.

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ z_s \\ 1 \end{bmatrix}$$

with $\alpha_x = fk_x$ and $\alpha_y = -fk_y$; and $x_{pix} = u'/w'$ and $y_{pix} = v'/w'$ so that:

3.

$$\begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = K[I_3 \,|\, 0_3]$$

where α_x and α_y are focal lengths in pixels and x_0 and y_0 are the coordinates of a center of the image in pixels, "s" is a skew parameter that accounts for roadside camera perspective angle and position, and $K[I_3|0_3]$ is the calibration matrix.

In yet another aspect of the present disclosure the method further includes adjusting the calibration matrix $K[I_3|0_3]$ based on known relative positions of the roadside camera and the range sensor, and based on a radial range or distance r_d between the roadside camera the object in the world coordinate system such that and a height h_1 of the roadside camera above ground at which the roadside camera is fixedly mounted, such that: $r = \sqrt{r_d^2 - h_1^2}$.

In yet another aspect of the present disclosure a method for range sensor-based calibration of roadside cameras includes capturing image data with one or more roadside cameras, the roadside cameras fixedly mounted to infrastructure and having a first field of view of at least a portion of a roadway. The method further includes capturing range sensor data with one or more range sensors, the range sensors fixedly mounted to infrastructure and having a second field of view of at least a portion of a roadway. The first and second fields of view at least partially overlap. The method further includes executing programmatic control logic stored within a memory of one or more controllers in communication with the one or more roadside cameras and the one or more range sensors. Each of the one or more controllers has a processor, the memory, and one or more input/output (I/O) ports. The I/O ports are in communication with the one or more range sensors and with the one or more roadside cameras. The processor executes the programmatic control logic including control logic for: constraining the image data from the roadside camera to a sub-region or region of interest (ROI) within the first field of view in which the image data captured by the roadside camera is reliably accurate and precise, and for constraining the range sensor data from the range sensor to a sub-region or ROI within the second field of view in which the range sensor data captured by the range sensor is reliably accurate and precise. The ROIs within the first and second fields of view are substantially identical. The method further includes executing control logic for constraining the image data to focus on a first movement of interest (MOI). The first MOI includes predictable and consistent motions of objects along the roadway. The method further includes executing control logic for constraining the range sensor data to focus on a second MOI. The second MOI includes predictable and consistent motions of objects along the roadway including motions in a predefined set of directions, linear or straight-line motions, and motions at predefined accelerations and velocities. The method further includes executing control logic for constraining the image data to define first objects of interest (OOIs) by selecting objects that satisfy a first ROI filter, and a first MOI filter, and that are directly and unobstructedly viewable by the roadside camera. The method further includes executing control logic for constraining, by a first OOI filter, the image data by selecting only first OOIs that are locations within the first ROI having predefined fixed or mobile positions, and constraining, by a second OOI filter, the range sensor data by selecting only second OOIs that are at locations within the second ROI having predefined fixed or mobile positions and that are directly and unobstructedly viewable by the range sensor. The predefined fixed or mobile positions include: locations of lane lines, positions of roadway signage, trees, traffic signals, telecommunications poles, fire hydrants, and/or curbs, and positions of movable or moving OOIs. The method further includes executing control logic for filtering, by a first position of interest (POI) filter, the image data to focus on a first POI, and filtering, by a second POI filter, the range sensor data to focus on a second POI. The method further includes executing control logic for selectively determining that objects detected by the roadside camera and by the range sensor satisfy both the first and second ROI, MOI, OOI, and POI filters and defining the objects as matching objects by comparing objects that satisfy both the first and second ROI, MOI, OOI, and POI filters and verifying one or more of the objects that satisfy both the first and second ROI, MOI, OOI, and POI filters is an identical object detected by both the roadside camera and the range sensor. The method further includes executing control logic for saving the matching objects in the memory for calibration, and calibrating the roadside camera by applying a correction factor based on the matching objects, so that the roadside camera accurately and precisely reports locations and movements of the matching objects for use in ADAS applications. The correction factor is based on known physical and geometrical locations of the roadside camera relative to known physical and geometrical locations of the range sensor, and takes a point location in a real-world coordinate system and translates the point to a camera coordinate system according to a calibration matrix $K[I_3|0_3]$ based on a world-homography matrix "H":

1.

$$H = KR[I_3 \,|\, -\tilde{C}]$$

where R is a rotation and C takes from the world coordinate system to the center of projection of the camera coordinate system;

2.

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ z_s \\ 1 \end{bmatrix}$$

with $\alpha_x = fk_x$ and $\alpha_y = -fk_y$; and $x_{pix} = u'/w'$ and $y_{pix} = v'/w'$ so that:

3.

$$\begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = K[I_3 \mid 0_3]$$

where α_x and α_y are focal lengths in pixels and x_0 and y_0 are the coordinates of a center of the image in pixels, "s" is a skew parameter that accounts for roadside camera perspective angle and position, and $K[I_3|0_3]$ is the calibration matrix.

In yet another aspect of the present disclosure the method further includes adjusting the calibration matrix $K[I_3|0_3]$ based on known relative positions of the roadside camera and the range sensor, and based on a radial range or distance r_d between the roadside camera the object in the world coordinate system such that and a height h_1 of the roadside camera above ground at which the roadside camera is fixedly mounted, such that: $r = \sqrt{r_d^2 - h_1^2}$.

Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

FIG. **1** is a schematic block diagram of a system for calibrating roadside cameras using range sensors for advanced driver assistance system (ADAS) applications according to an exemplary embodiment;

FIG. **2** is a diagram of a camera and a world reference frame for the system for calibrating roadside cameras using range sensors for ADAS applications of FIG. **1** according to an exemplary embodiment;

FIG. **3** is a diagram of a coordinate adjustment for the system for calibrating roadside cameras using range sensors for ADAS applications of FIGS. **1** and **2** according to an exemplary embodiment;

FIG. **4** is a perspective top view of a plurality of vignettes or frames defining aspects of the fields of view of the roadside cameras and range sensors of the system for calibrating roadside cameras using range sensors for ADAS applications of FIG. **1** according to an exemplary embodiment; and

FIG. **5** is a flowchart of a method of calibrating roadside cameras using range sensors for ADAS applications according to an exemplary embodiment.

DETAILED DESCRIPTION

The following detailed description is merely exemplary in nature and is not intended to limit the application and uses. Furthermore, there is no intention to be bound by expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. As used herein, the term "module" refers to hardware, software, firmware, electronic control component, processing logic, and/or processor device, individually or in a combination thereof, including without limitation: application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality.

Embodiments of the present disclosure may be described herein in terms of functional and/or logical block components and various processing steps. It should be appreciated that such block components may be realized by a number of hardware, software, and/or firmware components configured to perform the specified functions. For example, an embodiment of the present disclosure may employ various integrated circuit components, e.g., memory elements, digital signal processing elements, logic elements, look-up tables, or the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with a number of systems, and that the systems described herein are merely exemplary embodiments of the present disclosure.

For the sake of brevity, techniques related to signal processing, data fusion, signaling, control, and other functional aspects of the systems (and the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent example functional relationships and/or physical couplings between the various elements. It should be noted that alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure.

Referring now to FIG. **1** a system **10** for performing automatic calibration of roadside cameras **12** using range sensors **14** is shown schematically. The system **10** calibrates roadside cameras **12** by accurately and precisely ascertaining distances between the camera and data points or objects **16** in the real-world and also within a first field of view **18** of the roadside cameras **12**. While the objects **16** may be fixed objects or features such as trees, lamp poles, parking meters, traffic cones, curbs, lane lines, planter boxes, buildings, and the like, it should be appreciated that the objects **16** may be movable or moving objects or features such as humans, animals, vehicles **19**, or the like without departing from the scope or intent of the present disclosure. The system **10** further includes one or more range sensors **14**, each of the one or more range sensors **14** having a second field of view **24** that substantially overlaps the first field of view **18** of the roadside cameras **12**. The roadside cameras **12** and one or more sensors **14** are each in communication with one or more control modules or controllers **26**.

In several aspects, the roadside cameras **12** may include any of a variety of different types of cameras without departing from the scope or intent of the present disclosure. For example, the roadside cameras **12** may include cameras capable of capturing optical information in a variety of different wavelengths of light, including but not limited to: infrared, visible, ultraviolet, and the like. The roadside cameras **12** may also be capable of capturing a series of still image data, videographic data or a combination of still and videographic data. In several aspects, the roadside cameras **12** are disposed at one or more locations along a roadway **28**, including but not limited to fixed locations **30** proximate an intersection **32**. The fixed locations **30** may include infrastructure such as lamp poles, traffic signal supports, crosswalk signal supports, power poles, buildings, or in or on other such fixed locations **30**.

In further aspects, the sensors **14** may be any of a variety of different types of range sensor **14** or combinations of range sensors **14** without departing from the scope or intent of the present disclosure. For example, the sensors **14** may be additional cameras, Light Detection and Ranging (LiDAR) sensors, Radio Detection and Ranging (RADAR) sensors, Sound Navigation and Ranging (SONAR) sensors, ultrasonic sensors, or combinations thereof. Further, the range sensors **14** may have the ability to communicate with a Global Positioning System (GPS) in order to more accurately and precisely set a location of the range sensors **14** for use in calibrating the roadside cameras **12** as will be described in further detail herein.

Each of the controllers **26** is a non-generalized electronic control device having a preprogrammed digital computer or processor **34**, non-transitory computer readable medium or memory **36** used to store data such as control logic, software applications, instructions, computer code, data, lookup tables, and the like, and one or more input/output (I/O) ports **38**. Computer readable medium or memory **36** includes any type of medium capable of being accessed by a computer, such as read-only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), solid-state memory, or any other type of memory. A "non-transitory" computer readable medium or memory **36** excludes wireless, optical, or other communication link that transport electrical or other signals. A non-transitory computer readable medium or memory **36** includes media where data can be permanently stored and media where data can be stored and later overwritten, such as any type of program code, including source code, object code, and executable code. The processor **34** is configured to execute the code or instructions. In some examples, the controller **26** may be a dedicated wireless or Wi-Fi controller. The I/O ports **38** are configured to communicate through wired or wireless means using Wi-Fi protocols under IEEE 802.11x, Bluetooth communications protocols, radio frequency (RF) protocols, or the like.

In several aspects, the controllers **26** include one or more applications **40**. An application **40** is a software program configured to perform specific functions or sets of functions. The application **40** may include one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The applications **40** may be stored within the memory **36** or in an additional or separate memory **36**. Examples of the applications **40** include audio or video streaming services, audio or visual processing services, roadside camera **12** calibration application **40**, and the like.

Roadside cameras **12**, even when disposed on a fixed structure as described herein, are calibrated in order to operate correctly and provide accurate and precise position physical location information of objects **16** to advanced driver assistance systems (ADAS) in communication with the roadside cameras **12**. Calibration is particularly important for newly-installed roadside cameras **12**, and roadside cameras **12** whose physical position has changed. In some examples, the physical, geometrical position of a roadside camera **12** may change due to weather (high wind speed, snow, hail, or the like); vehicle **19** collisions with infrastructure-mounted roadside cameras **12**, construction, roadway **28** modifications, and the like. For roadside cameras **12** whose physical, geometrical position has changed, the positions of objects **16** within the first field of view **18** may be inaccurately reported. Accordingly, to correct for changes in physical positions of roadside cameras **12**, especially for roadside cameras **12** in use in ADAS processes, a calibration procedure is carried out by conducting a survey to determine the physical placement of objects **16** relative to the roadside cameras **12** and within the first field of view **18** of the roadside cameras **12**.

Turning now to FIG. **2** and with continuing reference to FIG. **1**, coordinate systems used to calibrate the roadside cameras **12** using the system **10** of the present disclosure are shown in further detail. Specifically, a camera coordinate system **50** and a world coordinate system **52** are shown in further detail. The camera coordinate system **50** is depicted in an exemplary, but non-limiting pose that corresponds to the physical location and angles of the first field of view **18**. The camera coordinate system **50** is applied in an image plane **54** with a plurality of image points **56** determined therein. Each of the plurality of image points **56** is defined by coordinates in the camera coordinate system **50** such that x_i is an X-location, y_i is a Y-location, and f is the focal distance from the roadside camera **12**. That is f is, in effect, the Z-location of the image points **56**. In several aspects, the image points **56** correspond to the physical locations of objects **16** in the real-world.

The world coordinate system **52** includes the physical locations of the objects **16**, where X_s is the X-location, Y_s is the Y-location, and Z_s is the Z-location defining one of the objects **16** in the world coordinate system **52**. In several aspects, a transformation is carried out between the camera coordinate system **50** and the world coordinate system **52** to generate a World-Camera Homography Matrix "H" that takes a point location in the real-world coordinate system **52** and translates the point location to the camera coordinate system **50** according to:

1.

$$H = KR[I_3 \mid -\tilde{C}]$$

where R is a rotation and C takes from the world coordinate system **52** to the center of projection of the camera coordinate system **50**.

2.

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ z_s \\ 1 \end{bmatrix}$$

with $\alpha_x = fk_x$ and $\alpha_y = fk_y$; and $x_{pix} = u'/w'$ and $y_{pix} = v'/w'$ so that:

3.

$$\begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = K[I_3 \mid 0_3]$$

where α_x and α_y are focal lengths in pixels and x_0 and y_0 are the coordinates of a center of the image in pixels, "s" is a skew parameter that accounts for roadside camera **12** perspective angle and position, and $K[I_3|0_3]$ is a calibration matrix.

Turning now to FIG. **3** and with continuing reference to FIGS. **1** and **2**, world coordinates of a range sensor **14** and roadside camera **12** are known and shown schematically as C_0 (for the world coordinates of the roadside camera **12**) and R_0 (for the world coordinates of the range sensor **14**). In examples where the roadside camera **12** and range sensor **14** are co-located, it should be appreciated that $C_0=R_0$. The range sensor **14** is used to detect a radial range of objects in the second field of view **24** and compute coordinates in the world coordinate system **52** using a radius r and the world coordinates R_0 of the range sensor **14**. In several aspects, the radius r is a horizontal or lateral distance from the fixed location of the roadside camera **12** to an object **16** at a known position in the world coordinate system **52**.

A vertical height h_1 of the roadside camera **12** defines a height above the ground at which the roadside camera **12** is fixedly mounted. Applying the Pythagorean Theorem, h_1 defines one adjacent side of a right triangle and r defines the second adjacent side of the right triangle, leaving r_d as the hypotenuse of the right triangle. In several aspects, r_d is a radial range or radial distance between the roadside camera **12** and the object **16** in the world coordinate system **52**. Accordingly, it should be appreciated that $r_d^2=r^2+h_1^2$ and therefore, $r=\sqrt{r_d^2-h_1^2}$.

In examples in which the range sensor **14** and roadside camera **12** are co-located, the radius r, the height h_1, and the r_d radial range or radial distance is the same for each of the range sensor **14** and roadside camera **12**. However, it should be appreciated that the range sensor **14** and roadside camera **12** are not required to be co-located. In several aspects, the range sensor **14** may be located proximate to the roadside camera **12** or may be displaced by a substantial distance from the roadside camera **12** without departing from the scope or intent of the present disclosure. That is, in some examples, the range sensor **14** and roadside camera **12** may be located at the same corner of an intersection **32** and/or fixedly mounted to the same physical infrastructure component and at, or proximate to, the same physical location the physical infrastructure component. In further examples, the range sensor **14** and roadside camera **12** may be located at opposite corners of an intersection **32**, at disparate heights, or the like. When the range sensor **14** and road are not co-located, i.e. when the range sensor **14** and roadside camera **12** are displaced by any distance, a mathematical transformation may be used to align the first and second fields of view **18**, **24** so that objects **16** detected in each of the first and second fields of view **18**, **24** are located in consistently by the range sensor **14** and roadside camera **12** of the system **10**.

In order to accurately calibrate the roadside camera **12**, the system **10** executes control logic stored in the memory **36** to transform data captured by one or more of the roadside camera **12** and the range sensor **14** via the World-Camera Homography Matrix "H" to find matching objects **16** within the first and second fields of view **18**, **24**. Representative coordinates of matched objects **16** in the camera coordinate system **50** and corresponding computed coordinates in the world coordinate system **52** are then used to calibrate the system **10**.

It will be appreciated that roadside cameras **12** and range sensors **14** such as RADARs and LiDARS utilize different mechanisms to sense or "see" objects **16**. Accordingly, the data corresponding to a given object **16** and obtained by each different sensor **14** or roadside camera **12** may not be a direct match to the data obtained by other sensors **14** or roadside cameras **12**. However, this limitation is overcome by leveraging contextual information, such as roadway **28** geometry and focusing only on a subset of all possible scenarios to simplify the dataset and increase the probability of finding matched objects **16**. Accordingly, the system **10** executes control logic to decrease the quantity of data points or objects **16** detected so that matching procedures are only carried out for a subset of all detected objects **16**.

Turning now to FIG. **4** and with continuing reference to FIGS. **1-3**, several vignettes or frames **100**, **102**, and **104** are shown. Based on roadway **28** geometry and known allowed vehicle **19** maneuvers, a region of interest (ROI), a motion of interest (MOI), an object of interest (OOI), and a relative position of interest (POI) are specified as constraints in both the first and second fields of view **18**, **24** of the roadside camera **12** and range sensor **14**, respectively.

The ROI is a predefined subset or sub-region of the first field of view **18** and the second field of view **24** that includes data in which the system **10** has high confidence that objects **16** within the ROI are likely to aid and not hinder the roadside camera **12** calibration process. In several aspects, the ROI includes a region in which the roadside camera **12** and the range sensor **14** have high accuracy and precision, such that objects **16** detected therein are in well-defined positions with well-defined motion characteristics. It will be appreciated that the ROI is a region extending from the roadside camera **12** position and from the range sensor **14** position to a predetermined, high-accuracy, high-precision distance therefrom. Beyond the predetermined, high-accuracy, high-precision distance, objects **16** detected are more likely to be obscured, occluded, or otherwise more inaccurately detected due to optical interference, electromagnetic interference, roadside camera **12** focal length, and other such factors.

Similarly, MOI includes movement of objects **16** within the first and second fields of view **18**, **24** that is highly predictable, and consistent, and that are unlikely to adversely effect the roadside camera **12** calibration process. That is, MOI may include, but is not necessarily limited to: objects **16** moving in a straight-line motion, or in a specific predefined direction, or the like. By contrast, vehicles **19** moving erratically, performing multiple lane changes, accelerating or decelerating abruptly, or executing other such actions are unlikely to result in good data for calibration purposes. Accordingly, such erratically-moving vehicles or other such objects **16** do not define MOIs.

OOIs include objects **16** within the first and second fields of view **18**, **24** that are easily viewed by the roadside camera **12** and by the range sensor **14** without being obscured, obfuscated, or the like. That is, the OOIs are objects **16** that are not obscured or partially obscured by other objects **16**, by infrastructure, by inclement weather conditions, or by other optical, electromagnetic, or audio impairments or conditions. Because objects **16** that are obscured or partially obscured are less "visible" to the roadside camera **12** and/or to the range sensors **14**, such obscured or partially obscured objects **16** are less well defined in the data captured by the roadside camera **12** and/or range sensors **14**, and accordingly, the data relating to such obscured or partially obscured objects **16** is less reliable than similar data relating to unobscured objects **16** for roadside camera **12** calibration.

POIs include positions of objects **16** that are of particular interest to the roadside camera **12** calibration process. Such

POIs may include locations within the ROI such as lane lines, the positions of roadway **28** signage, trees, traffic signals, telecommunications poles, fire hydrants, curbs, or other such objects **16** having well-defined positions, or the like. Such POIs offer well-defined and easily documented and reliable data for comparison between the roadside camera **12** data and the range sensor **14** data for calibration purposes. It should be appreciated, however, that while the POIs have been described hereinabove as generally being immobile positions of infrastructure, other movable positions may equally be POIs without departing from the scope or intent of the present disclosure. For example, the POIs may include the positions of objects **16** detectable within the first and second fields of view **18**, **24** by both the roadside camera **12** and the range sensor **14** and which have been identified as being OOIs within the ROIs, and/or having MOIs as well.

In several aspects, based on the ROI, MOI, OOI, and POI, the system **10** focuses on a specific segment of the roadway **28**, specific objects **16** (e.g., cars only, etc.), and the objects **16** are localized in simple terms. For example, the simplified terms may be the left and right sides of a two-lane roadway **28**. To further simplify, the system **10** executes control logic that identifies and matches left and right moving objects **16**. The system **10** then filters complicated scenarios by specifying motion constraints. In some examples, the motion constraints may include, but are not limited to: straight-line motion only, motion in a specific direction only, discard data containing lane changes, or the like. However, other constraints may be used without departing from the scope or intent of the present disclosure. Furthermore, to avoid occlusion-related issues whereby an object **16** in the foreground obscures or partially obscures an object **16** behind it, the system **10** focuses only on objects **16** in "front" of, or within the ROI. That is, the system **10** focuses on objects **16** having no other objects **16** in front of themselves. In more complicated scenarios, such as roadways **28** having multiple lanes of traffic moving in more than two directions at a given intersection **32**, advanced algorithms such as Bipartie graphs or Hungarian matching algorithms are leveraged to increase the accuracy and precision of object **16** matching in data from multiple sensor sources, including data from the roadside camera **12** and the range sensors **14**.

Referring again more specifically to the first, second and third frames **100**, **102**, and **104**, the first frame **100** is a depiction of a roadway **28** having multiple first objects **16'** disposed thereon. Each of the first objects **16'** is in motion in accordance with directional arrows **106**. A second set of objects **16"** are shown as well. The second set of objects **16"** are behind the first objects **16**, relative to the direction of motion **106** of the first objects **16'**. It should be appreciated that there is nothing in front of the first objects **16'** and that the first objects **16'** are moving generally towards a roadside camera **12** and a range sensor **14**.

The second frame **102** is a depiction of the first field of view **18** of the roadside camera **12** and contains road markings and the first objects **16'** but in a skewed perspective that corresponds to the roadside camera **12** pose relative to the real-world and the world coordinate system **52**. Likewise, the third frame **104** is a depiction of the second field of view **24** of the range sensor **14**. In the example shown in the third frame **104**, the range sensor **14** is a RADAR sensor and while the RADAR sensor has detected the same first objects **16'**, they are depicted as very small dots **108** that define the radar cross-sections of the first objects **16'**. Accordingly, the data detected by each of the roadside camera **12** and the RADAR sensor (range sensor **14**) are distinct from one another, but similarly localized.

It will be appreciated that object **16** classification in roadside camera **12** data versus radar cross section (RCS) data in RADAR sensor data can be leveraged to confirm object **16** matches. For example, the RCS of a truck is likely to be significantly larger than the RCS of a car. Likewise a car can be differentiated from a truck in roadside camera **12** data. Combined, the information from both the roadside camera **12** and range sensors **14**, such as RADAR sensors increases accuracy and precision in not only localizing objects **16** within the first and second fields of view **18**, **24**, but also the motions of such objects **16** and confirming object **16** matches.

Turning now to FIG. **5** and with continuing reference to FIGS. **1-4**, a flowchart depicting a method **200** of calibrating roadside cameras **12** with the system **10** of the present disclosure is shown. The method begins at block **202**. At block **204**, the system **10** executes control logic that causes the roadside camera **12** to obtain image and/or videographic data of the first field of view **18**. At block **206**, the system **10** executes control logic that applies a first ROI filter to the image and/or videographic data from the roadside camera **12**. The first ROI filter constrains the image and/or videographic data from the roadside camera **12** to the predefined sub-region or ROI of the first field of view **18**. The method **200** then proceeds to block **208**.

At block **208**, the system **10** executes control logic that applies a first MOI filter to the image and/or videographic data within the ROI of the first field of view **18**. The first MOI filter further constrains the data contained within the image and/or videographic data by selecting only the motions of objects **16** that are highly predictable, and consistent. In some examples, the first MOI filter selects movement detected in a straight-line, or in a specific predefined direction, or the like. The method **200** then proceeds to block **210**.

At block **210**, the system **10** executes control logic that applies a first OOI filter to the image and/or videographic data within the ROI and corresponding to the MOI filtered-data. The first OOI filter further constrains the data within the image and/or videographic data by selecting only objects **16** within the ROI, which have MOI, and that are easily viewed by the roadside camera **12** without being obscured, obfuscated, or the like. That is, the OOIs are objects **16** that are not obscured or partially obscured by other objects **16**, by infrastructure, by inclement weather conditions, or by other optical, electromagnetic, or audio impairments or conditions. The method **200** then proceeds to block **212**.

At block **212**, the system **10** executes control logic that applies a first POI filter or relative positioning filter to the image and/or videographic data from the roadside camera **12**. The first POI filter further constrains the data within the image and/or videographic data by selecting only objects **16** within the ROI, which have MOI, and that have been identified as OOIs and further identifies the positions of objects **16** that are of particular interest to the roadside camera **12** calibration process. Such POIs may include locations within the ROI such as lane lines, the positions of roadway **28** signage, trees, traffic signals, telecommunications poles, fire hydrants, curbs, or other such objects **16** having well-defined positions, or the like. Such POIs offer well-defined and easily documented and reliable data for comparison between the roadside camera **12** data and the range sensor **14** data for calibration purposes.

At block **214**, the system **10** executes control logic that causes the range sensor **14** to obtain range sensor data of the second field of view **24**. At block **216**, the system **10** executes control logic that applies a second ROI filter to the range sensor **14** data. The second ROI filter constrains the range sensor **14** data to a predefined sub-region or ROI of the second field of view **24**. The method **200** then proceeds to block **218**.

At block **218**, the system **10** executes control logic that applies a second MOI filter to the range sensor **14** data within the ROI of the second field of view **24**. The second MOI filter further constrains the data contained within the range sensor **14** data by selecting only the motions of objects **16** that are highly predictable, and consistent. In some examples, the second MOI filter selects movement detected in a straight-line, or in a specific predefined direction, or the like. The method **200** then proceeds to block **220**.

At block **220**, the system **10** executes control logic that applies a second OOI filter to the range sensor **14** data within the ROI and corresponding to the MOI filtered-data. The second OOI filter further constrains the data within the range sensor **14** data by selecting only objects **16** within the ROI, which have MOI, and that are easily viewed by the roadside camera **12** without being obscured, obfuscated, or the like. That is, the OOIs are objects **16** that are not obscured or partially obscured by other objects **16**, by infrastructure, by inclement weather conditions, or by other optical, electromagnetic, or audio impairments or conditions. The method **200** then proceeds to block **222**.

At block **222**, the system **10** executes control logic that applies a second POI filter or relative positioning filter to the range sensor **14** data. The second POI filter further constrains the data within the range sensor **14** data by selecting only objects **16** within the ROI, which have MOI, and that have been identified as OOIs and further identifies the positions of objects **16** that are of particular interest to the roadside camera **12** calibration process. Such POIs may include locations within the ROI such as lane lines, the positions of roadway **28** signage, trees, traffic signals, telecommunications poles, fire hydrants, curbs, or other such objects **16** having well-defined positions, or the like. Such POIs offer well-defined and easily documented and reliable data for comparison between the roadside camera **12** data and the range sensor **14** data for calibration purposes. It should be appreciated that while blocks **204-222** have been described sequentially herein, the first and second ROI filters may be executed in parallel, the first and second MOI filters may be executed in parallel, the first and second OOI filters may be executed in parallel, and the first and second POI filters may be executed in parallel without departing from the scope or intent of the present disclosure. Likewise, the first and second ROI, MOI, OOI, and POI filters may be executed in differing orders than those described herein, or may be executed entirely simultaneously and/or in parallel without departing from the scope or intent of the present disclosure.

At block **224** the system **10** executes matching control logic that determines whether objects **16** identified as having satisfied both the first ROI, MOI, OOI, and POI filters and the second ROI, MOI, OOI, and POI filters have been detected. Upon determining that objects **16** have been detected that satisfy both first and second ROI, MOI, OOI, and POI filters, the method **200** proceeds to block **226** where one or more objects **16** that have satisfied one or more of the first ROI, MOI, OOI, and POI filters, and which have also satisfied one or more of the second ROI, MOI, OOI, and POI filters are stored in the memory **36** for calibration of the roadside camera **12**. In several aspects, in order to properly calibrate the roadside camera **12**, the first and second ROI, MOI, OOI, and POI filters are satisfied by objects **16** detected at a plurality of different radial ranges or radial distances r_d from each of the roadside camera **12** and the range sensor **14**. The plurality of different radial ranges or radial distances r_d from each of the roadside camera **12** and the range sensor **14** improve the quality and accuracy of the calibration by avoiding the potential for calibrating or, in essence, focusing the calibration at a first radial range or distance r_d, while ignoring other possible radial ranges or distances r_d at which the calibration for the first radial range or distance r_d is, in fact, incorrect. Accordingly, the accuracy of the system **10** is improved by including objects **16** which have satisfied the first and second ROI, MOI, OOI, and POI filters, and which are located at a variety of different radial ranges or distances r_d from the roadside camera **12** and from the range sensor **14**. In further examples, when the roadside camera **12** and range sensor **14** are not co-located, the physical size and shape of the objects **16** detected and satisfying the first and second ROI, MOI, OOI, and POI filters may cause uncertainty as to whether or not the objects **16** detected are, in fact, matching objects **16**.

Accordingly, in order to ensure proper matching, at least a portion of the matching control logic includes a time-synchronization algorithm or process. The time-synchronization algorithm or process may be any of a variety of processes that accounts for the physical relative locations, image or data capture frequencies or periodicities of the roadside camera **12** and range sensor **14**, and the like, as well as the directionality, accelerations and velocities of the objects **16** detected and satisfying the first and second ROI, MOI, OOI, and POI filters are, in fact, the same objects **16** detected in both the first and second ROI, MOI, OOI, and POI filters. In some examples, the time-synchronization algorithm may be executed within the controller **26**, or within a separate and distinct time server, or within a distinct time-server application, subroutine, or control logic within the controller **26**.

Once matching objects **16** have been determined, the method **200** proceeds to block **228** where the method **200** calibrates the roadside camera **12** by applying a correction factor to position information for each object **16** detected within the first field of view **18** and satisfying the first ROI, MOI, OOI, and POI filters. The correction factor may be a mathematical weight or other such multiplicative factor that accounts for the relative positions of the range sensor **14** and roadside camera **12** in relation to the matching objects **16**.

Referring once more to block **224**, when matching objects **16** are not found, the method **200** proceeds to block **230** where the objects **16** detected, but not otherwise satisfying the first and second ROI, MOI, and POI filters are discarded. From blocks **228** and **230**, the method **200** proceeds to block **232**, where the method **200** ends. From block **232**, the method **200** proceeds back to block **202** where the method **200** runs again.

It should be appreciated that the method **200** may be executed or run automatically, recursively, continuously, periodically, or upon the occurrence of a particular condition, such as manual initialization by a human without departing from the scope or intent of the present disclosure. It should further be noted that while the method **200** may be run as described above, in many examples, the method **200** is only executed while the range sensor **14** is being used to calibrate the roadside camera **12**. That is, in many instances, roadside cameras **12** need only to be calibrated occasionally, or upon the occurrence of certain conditions, such as when a newly-installed roadside camera **12** is being calibrated, or for roadside cameras **12** whose physical, geometrical position has changed due to weather (high wind speed, snow, hail, or the like); vehicle **19** collisions with infrastructure-mounted roadside cameras **12**, construction, roadway **28** modifications, or the like.

Further, while in the foregoing, the calibration process has been described as being used to calibrate the roadside camera **12**, it should be appreciated that substantially the same or similar systems **10** and methods **200** as those described herein may be used to calibrate range sensors **14**, range sensors **14** and cameras **12**, or the like, without departing from the scope or intent of the present disclosure.

A system **10** and method **200** of the present disclosure for calibrating roadside cameras **12** calibrating roadside cameras **12** with range sensors **14**, or vice versa, offers several advantages. These include, improved efficiency over manual calibration processes, decreased complexity, and reduction in time and effort needed to perform roadside camera **12** and/or range sensor **14** calibration for use in camera-based advanced driver assistance systems (ADAS) applications.

The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

What is claimed is:

1. A system for range sensor-based calibration of roadside cameras comprising:

one or more roadside cameras, the roadside cameras fixedly mounted to infrastructure and having a first field of view of at least a portion of a roadway;

one or more range sensors, the range sensors fixedly mounted to infrastructure and having a second field of view of at least a portion of a roadway, wherein the first and second fields of view at least partially overlap;

one or more controllers, each of the one or more controllers having a processor, a memory, and one or more input/output (I/O) ports, the I/O ports in communication with the one or more range sensors and the one or more roadside cameras; the memory storing programmatic control logic; the processor executing the programmatic control logic; the programmatic control logic including:

a first control logic for causing each of the one or more roadside cameras to capture image data, and for causing each of the range sensors to capture range sensor data;

a second control logic that filters the image data to focus on a first region of interest (ROI), and that filters the range sensor data to focus on a second ROI;

a third control logic that filters the image data to focus on a first movement of interest (MOI), and that filters the range sensor data to focus on a second MOI;

a fourth control logic that filters the image data to focus on a first object of interest (OOI), and that filters the range sensor data to focus on a second OOI;

a fifth control logic that filters the image data to focus on a first position of interest (POI), and that filters the range sensor data to focus on a second POI;

a sixth control logic that selectively determines that objects detected by the roadside camera and by the range sensor satisfy both the first and second ROI, MOI, OOI, and POI filters and defines the objects as matching objects;

a seventh control logic that saves the matching objects in the memory for calibration; and

an eighth control logic that calibrates the roadside camera by applying a correction factor based on the matching objects, so that the roadside camera accurately and precisely reports locations and movements of the matching objects for use in ADAS applications.

2. The system of claim **1**, wherein the second control logic further comprises:

control logic that constrains the image data from the roadside camera to a sub-region or ROI within the first field of view in which the image data captured by the roadside camera is reliably accurate and precise;

control logic that constrains the range sensor data from the range sensor to a sub-region or ROI within the second field of view in which the range sensor data captured by the range sensor is reliably accurate and precise; and

wherein the ROIs within the first and second fields of view are substantially identical.

3. The system of claim **1**, wherein the third control logic further comprises:

control logic that constrains the image data to focus on the first MOI, wherein the first MOI includes predictable and consistent motions of objects along the roadway; and

control logic that constrains the range sensor data to focus on the second MOI, wherein the second MOI includes predictable and consistent motions of objects along the roadway.

4. The system of claim **3**, wherein the predictable and consistent motions of objects along the roadway further comprise:

motions in a predefined set of directions, linear or straight-line motions, and motions at predefined accelerations and velocities.

5. The system of claim **1**, wherein the fourth control logic further comprises:

control logic that constrains the image data to define first OOIs by selecting objects that satisfy the first ROI filter, the first MOI filter, and that are directly and unobstructedly viewable by the roadside camera; and

control logic that constrains the range sensor data to define second OOIs by selecting objects that satisfy the second ROI filter, the second MOI filter, and that are directly and unobstructedly viewable by the range sensor.

6. The system of claim **5**, wherein the fifth control logic further comprises:

control logic that constrains the image data by selecting only first OOIs that are locations within the first ROI having predefined fixed or mobile positions;

control logic that constrains the range sensor data by selecting only second OOIs that are at locations within the second ROI having predefined fixed or mobile positions; and

wherein the predefined fixed or mobile positions comprise: locations of lane lines, positions of roadway signage, trees, traffic signals, telecommunications poles, fire hydrants, and/or curbs, and positions of movable or moving OOIs.

7. The system of claim **1**, wherein the sixth control logic further comprises:

control logic that defines the objects as matching objects by comparing objects that satisfy both the first and second ROI, MOI, OOI, and POI filters and verifying one or more of the objects that satisfy both the first and second ROI, MOI, OOI, and POI filters is an identical object detected by both the roadside camera and the range sensor.

8. The system of claim 7, wherein the eighth control logic further comprises:

control logic that applies the correction factor to image data captured by the roadside camera, wherein the correction factor is based on known physical and geometrical locations of the roadside camera relative to known physical and geometrical locations of the range sensor, and takes a point location in a real-world coordinate system and translates the point to a camera coordinate system according to a calibration matrix $K[I_3|0_3]$ is a calibration matrix based on a world-homography matrix "H":

1.

$$H = KR[I_3 \mid -\tilde{C}]$$

where R is a rotation and C takes from the world coordinate system to the center of projection of the camera coordinate system;

2.

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ z_s \\ 1 \end{bmatrix}$$

with $\alpha_x = fk_x$ and $\alpha_y = -fk_y$; and $x_{pix} = u'/w'$ and $y_{pix} = v'/w'$ so that:

3.

$$\begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = K[I_3 \mid 0_3]$$

where α_x and α_y are focal lengths in pixels and x_0 and y_0 are the coordinates of a center of the image in pixels, "s" is a skew parameter that accounts for roadside camera perspective angle and position, and $K[I_3|0_3]$ is a calibration matrix.

9. The system of claim 8, wherein the calibration matrix $K[I_3|0_3]$ is further adjusted based on the known relative positions of the roadside camera and the range sensor, and based on a radial range or distance r_d between the roadside camera the object in the world coordinate system such that and a height h_1 of the roadside camera above ground at which the roadside camera is fixedly mounted, such that: $r = \sqrt{r_d^2 - h_1^2}$.

10. A method for range sensor-based calibration of roadside cameras comprising:

capturing image data with one or more roadside cameras, the roadside cameras fixedly mounted to infrastructure and having a first field of view of at least a portion of a roadway;

capturing range sensor data with one or more range sensors, the range sensors fixedly mounted to infrastructure and having a second field of view of at least a portion of a roadway, wherein the first and second fields of view at least partially overlap;

executing programmatic control logic stored within a memory of one or more controllers in communication with the one or more roadside cameras and the one or more range sensors, each of the one or more controllers having a processor, the memory, and one or more input/output (I/O) ports, the I/O ports in communication with the one or more range sensors and the one or more roadside cameras; the processor executing the programmatic control logic; the programmatic control logic including:

filtering the image data to focus on a first region of interest (ROI), and filtering the range sensor data to focus on a second ROI;

filtering the image data to focus on a first movement of interest (MOI), and filtering the range sensor data to focus on a second MOI;

filtering the image data to focus on a first object of interest (OOI), and filtering the range sensor data to focus on a second OOI;

filtering the image data to focus on a first position of interest (POI), and filtering the range sensor data to focus on a second POI;

selectively determining that objects detected by the roadside camera and by the range sensor satisfy both the first and second ROI, MOI, OOI, and POI filters and defining the objects as matching objects;

saving the matching objects in the memory for calibration; and

calibrating the roadside camera by applying a correction factor based on the matching objects, so that the roadside camera accurately and precisely reports locations and movements of the matching objects for use in ADAS applications.

11. The method of claim 10, further comprising:

constraining the image data from the roadside camera to a sub-region or ROI within the first field of view in which the image data captured by the roadside camera is reliably accurate and precise;

constraining the range sensor data from the range sensor to a sub-region or ROI within the second field of view in which the range sensor data captured by the range sensor is reliably accurate and precise; and

wherein the ROIs within the first and second fields of view are substantially identical.

12. The method of claim 10, further comprising:

constraining the image data to focus on the first MOI, wherein the first MOI includes predictable and consistent motions of objects along the roadway; and

constraining the range sensor data to focus on the second MOI, wherein the second MOI includes predictable and consistent motions of objects along the roadway.

13. The method of claim 12, wherein constraining the image and sensor data further comprises:

selecting predictable and consistent motions of objects along the roadway including motions in a predefined set of directions, linear or straight-line motions, and motions at predefined accelerations and velocities.

14. The method of claim 10, further comprising:

constraining the image data to define first OOIs by selecting objects that satisfy the first ROI filter, the first MOI filter, and that are directly and unobstructedly viewable by the roadside camera; and

constraining the range sensor data to define second OOIs by selecting objects that satisfy the second ROI filter, the second MOI filter, and that are directly and unobstructedly viewable by the range sensor.

15. The method of claim 14, further comprising

constraining the image data by selecting only first OOIs that are locations within the first ROI having predefined fixed or mobile positions;

constraining the range sensor data by selecting only second OOIs that are at locations within the second ROI having predefined fixed or mobile positions; and

wherein the predefined fixed or mobile positions comprise: locations of lane lines, positions of roadway signage, trees, traffic signals, telecommunications poles, fire hydrants, and/or curbs, and positions of movable or moving OOIs.

16. The method of claim **10**, further comprising:

defining the objects as matching objects by comparing objects that satisfy both the first and second ROI, MOI, OOI, and POI filters and verifying one or more of the objects that satisfy both the first and second ROI, MOI, OOI, and POI filters is an identical object detected by both the roadside camera and the range sensor.

17. The method of claim **16**, further comprising:

applying the correction factor to image data captured by the roadside camera, wherein the correction factor is based on known physical and geometrical locations of the roadside camera relative to known physical and geometrical locations of the range sensor, and takes a point location in a real-world coordinate system and translates the point to a camera coordinate system according to a calibration matrix $K[I_3|0_3]$ is a calibration matrix based on a world-homography matrix "H":

1.

$$H = KR[I_3 \mid -\tilde{C}]$$

where R is a rotation and C takes from the world coordinate system to the center of projection of the camera coordinate system;

2.

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ z_s \\ 1 \end{bmatrix}$$

with $\alpha_x = fk_x$ and $\alpha_y = -fk_y$; and $x_{pix} = u'/w'$ and $y_{pix} = v'/w'$ so that:

3.

$$\begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = K[I_3 \mid 0_3]$$

where α_x and α_y are focal lengths in pixels and x_0 and y_0 are the coordinates of a center of the image in pixels, "s" is a skew parameter that accounts for roadside camera perspective angle and position, and $K[I_3|0_3]$ is a calibration matrix.

18. The method of claim **17**, further comprising:

adjusting the calibration matrix $K[I_3|0_3]$ based on known relative positions of the roadside camera and the range sensor, and based on a radial range or distance r_d between the roadside camera the object in the world coordinate system such that and a height h_1 of the roadside camera above ground at which the roadside camera is fixedly mounted, such that: $r=\sqrt{r_d^2-h_1^2}$.

19. A method for range sensor-based calibration of roadside cameras comprising:

capturing image data with one or more roadside cameras, the roadside cameras fixedly mounted to infrastructure and having a first field of view of at least a portion of a roadway;

capturing range sensor data with one or more range sensors, the range sensors fixedly mounted to infrastructure and having a second field of view of at least a portion of a roadway, wherein the first and second fields of view at least partially overlap;

executing programmatic control logic stored within a memory of one or more controllers in communication with the one or more roadside cameras and the one or more range sensors, each of the one or more controllers having a processor, the memory, and one or more input/output (I/O) ports, the I/O ports in communication with the one or more range sensors and the one or more roadside cameras; the processor executing the programmatic control logic; the programmatic control logic including:

constraining the image data from the roadside camera to a sub-region or region of interest (ROI) within the first field of view in which the image data captured by the roadside camera is reliably accurate and precise;

constraining the range sensor data from the range sensor to a sub-region or ROI within the second field of view in which the range sensor data captured by the range sensor is reliably accurate and precise, wherein the ROIs within the first and second fields of view are substantially identical;

constraining the image data to focus on a first movement of interest (MOI), wherein the first MOI includes predictable and consistent motions of objects along the roadway;

constraining the range sensor data to focus on a second MOI, wherein the second MOI includes predictable and consistent motions of objects along the roadway including motions in a predefined set of directions, linear or straight-line motions, and motions at predefined accelerations and velocities;

constraining the image data to define first objects of interest (OOIs) by selecting objects that satisfy a first ROI filter, a first MOI filter, and that are directly and unobstructedly viewable by the roadside camera;

constraining, by a first OOI filter, the image data by selecting only first OOIs that are locations within the first ROI having predefined fixed or mobile positions;

constraining, by a second OOI filter, the range sensor data by selecting only second OOIs that are at locations within the second ROI having predefined fixed or mobile positions and that are directly and unobstructedly viewable by the range sensor;

wherein the predefined fixed or mobile positions comprise: locations of lane lines, positions of roadway signage, trees, traffic signals, telecommunications poles, fire hydrants, and/or curbs, and positions of movable or moving OOIs;

filtering, by a first position of interest (POI) filter, the image data to focus on a first POI, and filtering, by a second POI filter, the range sensor data to focus on a second POI;

selectively determining that objects detected by the roadside camera and by the range sensor satisfy both the first and second ROI, MOI, OOI, and POI filters and defining the objects as matching objects by comparing objects that satisfy both the first and second ROI, MOI, OOI, and POI filters and verifying one or more of the objects that satisfy both the first and second ROI, MOI, OOI, and POI filters is an identical object detected by both the roadside camera and the range sensor;

saving the matching objects in the memory for calibration; and

calibrating the roadside camera by applying a correction factor based on the matching objects, so that the roadside camera accurately and precisely reports locations and movements of the matching objects for use in ADAS applications, wherein the correction factor is based on known physical and geometrical locations of the roadside camera relative to known physical and geometrical locations of the range sensor, and takes a point location in a real-world coordinate system and translates the point to a camera coordinate system according to a calibration matrix $K[I_3|0_3]$ based on a world-homography matrix "H":

$$H = KR\left[I_3 \mid -\tilde{C}\right] \tag{1}$$

where R is a rotation and C takes from the world coordinate system to the center of projection of the camera coordinate system;

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ z_s \\ 1 \end{bmatrix} \tag{2}$$

with $\alpha_x = fk_x$ and $\alpha_y = -fk_y$; and $x_{pix} = u'/w'$ and $y_{pix} = v'/w'$ so that:

$$\begin{bmatrix} \alpha_x & s & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = K[I_3 \mid 0_3] \tag{3}$$

where α_x and α_y are focal lengths in pixels and x_0 and y_0 are the coordinates of a center of the image in pixels, "s" is a skew parameter that accounts for roadside camera perspective angle and position, and $K[I_3|0_3]$ is the calibration matrix.

20. The method of claim **19**, further comprising:

adjusting the calibration matrix $K[I_3|0_3]$ based on known relative positions of the roadside camera and the range sensor, and based on a radial range or distance r_d between the roadside camera the object in the world coordinate system such that and a height h_1 of the roadside camera above ground at which the roadside camera is fixedly mounted, such that: $r=\sqrt{r_d^2-h_1^2}$.

* * * * *